

A discrete deterministic base for continuum physics

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Abstract

Many programs in fundamental physics reach the same structures from above. Gravity appears as the thermodynamics of entanglement, quantum theory as an information-processing reconstruction, spacetime as a discrete causal order, and in the conserved-current approach all of physics follows from one quantity whose only law is that it is neither created nor destroyed. This paper works from the bottom instead. The base is a single growing mesh of one substance carrying a ternary value in $\{-1, 0, +1\}$, moved by one local reversible charge-conserving rule, on the four-dimensional hyperbolic honeycomb $\{3, 4, 3, 4\}$ whose cells are 24-cells. A conserved current is built on it directly, and the central conjecture is that its coarse-grained form is a conserved-current field theory of the kind the continuum programs describe. We test this one result at a time and grade each as derived on the substrate, parallel by a different route, or open with a stated obstruction. The firm-tier results are exact and integer-checked. Charge conservation is the continuity law $\nabla \cdot J = 0$ on both a flat torus and the curved $\{3, 4, 3, 4\}$ mesh, the 24-cell direction set is fixed by a spinor test and supplies the isotropy a four-dimensional lattice gas requires, and one recoverability functional ties horizons, decoherence, and conservation together. The middle rungs run parallel, reaching a finite causal speed, entanglement that builds geometry, the Born rule and the Tsirelson bound from a deterministic base, and the wake as a measured expansion scalar. The open frontiers are stated with their tests. Exact Lorentz invariance on a fixed lattice, three chiral generations, the single measurement outcome, and a full field theory whose stress tensor sources curvature all remain unsolved. The whole comparison rests on a reproducible suite of deterministic, controlled experiments written as fully open source software [28]. This is exploratory, feasibility-stage work.

1. Opening

The Problem

Programs that begin in very different places keep arriving at the same small set of objects. Gravity comes out as the thermodynamics of horizons and the entanglement of the vacuum [46, 48]. Spacetime geometry comes out as the pattern of entanglement in a quantum state [36]. Quantum theory comes out as the only information-processing structure that meets a short list of operational axioms [16, 34]. Spacetime itself comes out as a discrete causal order [33], and the dynamics of matter come out of a reversible local rule [14, 15].

The conserved-current program of Chronoflux pushes this furthest [42]. There the whole of physics is forced from one quantity whose only law is that it is neither created nor destroyed. Inertia, relativity, gravity, the

quantum, and cosmic expansion are read as faces of a single conserved flow obeying a continuity equation. The discipline is that conservation comes first and geometry comes second.

Each of these programs starts in the continuum and posits the principle it needs, a thermodynamic identity, a Hilbert space, a conserved current, a causal set, then shows that the rest of physics is the only consistent completion. The principle is the input. What is missing in each case is a base underneath the principle. Why a conserved current. Why this much entanglement. Why a finite count of states in a bounded region. Each program takes its principle as given rather than explaining where it comes from.

This paper works in the opposite direction, from the base upward. It starts from a single discrete substance with no continuum, no randomness, and no fitted constant, and asks whether the structures the top-down programs reach are forced from below. The base is one growing mesh of a ternary value moved by one reversible, charge-conserving rule. The conjecture is that its coarse-grained face is a conserved-current field theory of the kind those programs describe.

We do not claim to merge the programs or to finish the derivation. We position the discrete base against the continuum programs in one frame and show, one result at a time, where the base derives a continuum result, where it runs parallel to a program by a different route, and where the bridge is an open conjecture with a stated falsifying test. The spine is the continuity of one conserved value, which holds exactly on the lattice. Everything else is graded against it.

The Claim

The full derivation of the base belongs to the theory paper [27]. What the rest of this paper rests on is stated here once and used throughout.

Vibe is a discrete, deterministic, reversible substrate that is conjectured to coarse-grain toward a conserved-current field theory of the kind the continuum programs describe, with no fitted continuum constant and no randomness. The same structures those programs reach from the top, vibe builds from the bottom. This is a conjecture with one firm anchor, not a finished derivation.

The model rests on two axioms. The first names the substance, the second names its motion.

Axiom 1 (The vibe) The one substance is the vibe, which is experience. A vibe's tone is the quality of that experience, not a thing apart from it. There is nothing beneath it.

Axiom 2 (The flow) The whole is the flow, the eight base elements running together as one, the vibe in endless motion.

The first axiom is the one place the model reaches past physics. Nothing downstream depends on the substance being felt. A reader who wants only the physics may read the vibe as a plain three-valued charge and lose nothing. We state it because the structure that yields the physics is, on the fuller reading, the structure of experience itself.

The base is exactly eight elements, no more. Three give the where, two the what, two the how, and one the when. Each is a four-letter word, so each doubles as a clean symbol.

element	role	what it is
mesh	where	the whole space, the growing $\{3, 4, 3, 4\}$ hyperbolic honeycomb of 24-cells
dock	where	one cell, a here-now where tones meet and move on, the 24-cell with twenty-four neighbors
site	where	one of a dock's twenty-four directions, a D_4 root, holding one tone
tone	what	the value at a site, one of $\{-1, 0, +1\}$
lean	what	the order $-1 < 0 < +1$, turning tone into signed charge and giving growth its direction
knit	how	the law of the state, collide then stream, once a beat, reversible and charge-conserving
wake	how	the law of the space, new docks born quiet at the growing edge, beat after beat
beat	when	one discrete tick of time

Table 1: The eight base elements. The whole model is these and nothing else.

Nothing is ever added underneath the eight. The base is discrete, deterministic, and carries no randomness. A ternary value, a finite direction set, a discrete beat, a rule that is a permutation, and a finite symmetry group. Real numbers appear only as measured outputs, never at the base. The continuum, Lie groups, the Dirac and Maxwell equations, Lorentz invariance, and the smooth metric are all emergent by coarse-graining. Vibes pass through one to three middle layers before they are physics, so a raw tone is not a particle and a first cluster is not a force.

This paper grades every claim on one scale, kept fixed throughout. **F**, firm, a structure-identical bridge established on the substrate. **P**, parallel, the same shape reached by an independent route but not yet derived from the base. **O**, open, a well-posed frontier with a stated falsifying test, unsolved on both sides. The result labels are narrower. **Derived** is forced from the base with no measured number fed in. **Parallel** is the same destination by a different road. **Open** is unsolved with a named obstruction. A separate axis runs alongside. Tier A is derived from the base, Tier B is checked against external numbers, and a Standard Model constant fed in for comparison never counts as a derivation. The two axes are read together and the substrate status governs.

The Method

Every result in this paper is an in-silico computation on the substrate, never a measurement against nature. The protocol is fixed and the same for all experiments, and it is designed so that a result is the gap between a run and a control rather than a number read off in isolation. The whole comparison rests on a reproducible suite [28].

The base rule is deterministic, with no randomness anywhere. It is a reversible permutation of the tones. Initial conditions are deterministic patterns or the vacuum, never random seeds. Scale is varied by changing the lattice size, not by averaging over draws. A run is bit-identical on two machines. This matters for the claims. A deterministic base that still yields the Born rule and the Tsirelson bound is a stronger result than one that feeds in quantum randomness by hand.

Every claim is paired with a control that could fail. If the result were an artifact, the control would expose it. Most often it is the lossy, erasing rule run against the conserving one, or a flat substrate run against the curved one. A result counts only as the difference between the run and its control. Where the conserving rule gives an exact zero, the lossy control gives a nonzero number equal to the charge it destroys.

Thresholds are fixed before running. Pass conditions are set in advance, and where possible they are exact integer or ratio equalities with no tolerance, so a result is an identity rather than a fit. The discipline borrowed from Chronoflux is one substrate-parameter assignment that must fit every arena at once, with the pass bars set before the data are seen.

Each experiment carries a stable code of the form E-XXX-NNNN, registered once in a catalog. The three letters mark the arena, GRV for gravity, RLT for relativity, SPN for spin, FLD for fluid, CSM for cosmology, MTH for method, FND for foundations, and so on. The four digits are the registry number. Each result is also given a depth grade. L3 is an emergent and novel result from the base rule with a control. L2 is known physics reproduced on the substrate. L1 is known math confirmed. L0 is a consistency note, kept and labeled as circular rather than dropped.

The evidence base is 615 experiments across 18 categories, 84 at L3, 389 at L2, 128 at L1, and 14 at L0, with 412 backing a specific claim in the papers. The anchors of this paper are E-FND-0030, E-GRV-0039, E-GRV-0042, E-GRV-0040, E-GRV-0041, and E-MTH-0001, alongside the ladder-push results E-RLT-0039, E-CSM-0043, E-FLD-0009, and E-FLD-0010.

Negative results are reported as findings. A push that turns out not to be cleanly buildable is recorded as such. Two negatives from this round are reported here. A metric from the bare tone correlations does not build, because the ternary tone field is short-range and carries no extended structure to reconstruct a metric from. A clean geometry-dependent settling time does not build, because the observable is too noisy. These count as results.

2. The Comparison

The Current

This is the foundation of the whole bridge, and it is exact and concrete. Everything else in the paper is graded against it.

Every dock carries twenty-four tones, one per direction. From them, build two things. The charge density is the scalar tone of a dock, the plain sum of its twenty-four tones. The current vector weights each tone by the D_4 root 4-vector of its direction and sums, giving a genuine 4-vector at every dock. So every dock already carries a 4-current, a density and a flow, with no coarse-graining, no fitting, and no new field. This is vibe's tone-current.

The rule only moves it. Collide rotates tone among a dock's head-on direction-pairs, which conserves the dock's charge. Stream carries each tone to the neighbor its direction points at, which permutes tones across docks. Both steps are exact and reversible. The total charge over the whole mesh is invariant under the rule, beat after beat, true microscopically by construction. E-FND-0030 confirms it directly, with a lossy control that fails. Tier F, Derived.

The continuity law follows. Take any block of docks. Over one beat, the change in charge inside the block equals minus the net flux of the current across its boundary. With the conserving rule this holds with residual exactly zero. This is the discrete divergence theorem. Block-average over growing blocks and the per-block sum rule becomes the smooth field equation, the time change of density plus the divergence of the

current equal to zero. That is $\nabla \cdot J = 0$ written for vibe’s block-averaged current, the central primitive of the conserved-current program reached from the discrete side.

This rung is where vibe and several continuum programs meet, each from the top. The conserved-current program of Herbert is one conserved four-dimensional current whose only law is zero divergence, with relativity, gravity, the quantum, and cosmology forced from it as a recovery ladder [42]. That is vibe’s spine reached from the opposite end. An independent worker reasoning purely in the continuum arrived at conservation-first plus a recovery ladder, which validates vibe’s choice of spine. The tension is that his fitted continuum constants and his astrophysical predictions do not import, and his one-way cosmological leg conflicts with vibe’s one-way wake.

The deeper move these programs share with vibe is to make conservation logically prior to geometry. The computational-universe and it-from-bit programs assert the same base posture, that physics is one informational rule [44]. Vibe realizes that posture concretely as a conserved ternary current rather than leaving it a slogan. The rule conserves first, and the structure that follows is forced.

The Programs

Vibe is measured against a family of continuum programs, not a single one. They differ in starting point but share a posture. Each puts one principle first and shows that physics is the only consistent completion. This section lays them out in one frame so the rungs that follow have clear targets. Each program is given one of three relations to vibe. A validation, where the program supports vibe or vibe confirms it. A tension, where a stronger claim fails or a no-go binds. A parallel, where the same shape is reached by a different route.

The conserved-current program comes first, because its spine is vibe’s spine. Chronoflux is its one developed instance [42]. It takes one conserved four-dimensional current whose only law is zero divergence and climbs a recovery ladder to inertia, relativity, gravity, the quantum, and cosmology, with a consistency matrix that demands one parameter assignment fit every arena at once. Vibe builds that same current from the discrete side, so the two run the same program from opposite ends. The relation is firm at the bottom rung, and above it the program is a target. Chronoflux is single-author, unreviewed, and unreplicated, so we treat its structure and its discipline as serious and its specific predictions as unconfirmed. It is one program among several, not the subject of the paper.

The gravity-from-thermodynamics programs supply the next family. Einstein’s equation comes out as a thermodynamic equation of state from horizon entropy and vacuum entanglement equilibrium [46, 47], and as the counting of surface degrees of freedom with the cosmological constant left as an integration constant [48]. Both derivations need a finite count of states in a bounded region, which vibe’s discreteness supplies for free with no tuning. These are firm fits. Entropic gravity and its emergent dark sector are the closest published cousin of vibe’s wake [11], but the derivation is contested, so it is inspiration and parallel, not a settled rung, and its warning that the law cannot depend on mass alone is a real constraint vibe must respect.

The entanglement-geometry program states that spacetime connectivity is entanglement, with geometry pinching off as the entanglement between regions is cut [36]. Vibe reproduces the co-variation on the substrate, where two regions entangle only through a geometric mediator and the mutual information dies when the

channel is severed. The relation is firm on the shape and parallel on the route, since the substrate result uses free-fermion proxies on a graph rather than entanglement generated by the ternary rule itself.

The quantum reconstruction programs set the checklist the emergent quantum surface must pass. Quantum theory is recovered as the unique structure meeting operational axioms of purification and information processing [16], or from a short list of reasonable axioms on outcomes [34, 31], with the toy-model line marking the boundary between quantum and classical statistics [41]. Vibe reaches the Born rule and the Tsirelson bound by a discrete, deterministic route, which is a parallel demonstration, not a derivation of the continuum reconstruction. The superdeterminism defense is vibe’s own stance, framed statistically [43]. The natively non-classical correlation shape is kept from the work of Christian while its local-realism claim is rejected [25].

The causal-set and discrete-gravity programs build or constrain emergent spacetime. The causal-set program reaches exact Lorentz invariance only by random sprinkling of points [33, 12], the very move vibe forbids, which makes Lorentz on a fixed lattice vibe’s sharpest open wall. Causal dynamical triangulations give a discrete-to-continuum route to four-dimensional spacetime with dimensional reduction at small scales [20], relational and graph-based quantum gravity supply the relational stance [3, 49], and causal invariance reappears as general covariance in the rewriting-system models [24]. These are parallels with one shared open tension at the lattice.

The discrete-rule programs put a local reversible rule directly under physics. The cellular-automaton interpretation derives quantum behavior from a deterministic rule and names the two hard walls of Lorentz and locality [14]. The Dirac equation is recovered from a quantum cellular automaton with a doubly-special route to Lorentz [15], and the same equation falls out of discrete sign-alternation and discrete walks [32, 13, 30]. Reversible conservative logic and the lattice-gas tradition are the direct ancestors of vibe’s rule, and four-dimensional isotropy is exactly what forces the 24-cell coin [9, 50, 8, 21]. The hyperbolic-honeycomb substrate, its cell-addressing, and the idea of computing on it draw on Margenstern’s two volumes [37, 38]. These are firm program-level kin, with the same open walls.

Two further families bracket the climb. The exceptional-algebra programs reach D_4 , the 24-cell, the octonions, and the gauge group from the top [22, 39, 6], on the algebraic ceiling of the four normed division algebras [2] and the geometry of the 24-cell and its symmetry [17, 23]. Lisi’s E_8 proposal is the most ambitious of these, fitting every known particle into the single exceptional group E_8 [1]. Vibe builds the same nucleus from the bottom. The shared open wall is triality to three chiral generations, bound by the Distler-Garibaldi no-go, which proved that no faithful three-generation structure embeds in E_8 the way the proposal needs [19], a wall that also limits the grand-unified and left-right routes [18, 29]. The upper-layer programs supply the tools for self and scale. Causal emergence gives the test and the lossy-middle constraint [10], the Markov-blanket and observer-boundary work gives the self [26, 5], discrete scale relativity gives the self-similar tower [40], and constructor theory supplies the method and the gravity-induced-entanglement witness [7, 4, 45].

This recurring pattern is the paper’s underlying argument. Independent workers from string holography, division algebras, cellular automata, causal sets, and quantum foundations keep arriving at the same small set of objects. One conserved local primitive, the exceptional algebra, geometry from entanglement, and a finite state count. Vibe supplies the one thing each of them assumes, a concrete discrete deterministic base from which these objects are derived rather than posited.

The Ladder

The continuum programs climb from the conserved current to all of physics, each rung forced as the only lawful completion of the one below. Vibe climbs the same ladder from the discrete side. We state each rung and mark vibe’s status, shown on the substrate, parallel, or open, with the experiment that backs it.

rung	continuum forces	vibe mechanism	status	experiment
conserved current continuity	one primitive, $\nabla \cdot J = 0$ zero divergence	tone-current from the 24 sites, conserved by the rule exact discrete closure at every block scale on the flat torus, and at every region on the curved $\{3, 4, 3, 4\}$ mesh	Derived, F Derived, flat and curved, F	E-FND-0030, E-GRV-0039 E-GRV-0039, E-GRV-0042
finite speed to relativity	special relativity	ballistic cone, conserved momentum current, sharp speed one, doubly-special dispersion	Parallel, full Lorentz open	E-RLT-0010, E-RLT-0039, E-RLT-0018
variation to geometry	Riemannian geometry	metric from a coarse field via the gapless free-fermion proxy, the bare tone is short-range	Parallel via proxy	E-GRV-0012
stress to curvature	Einstein’s equation	area-law gravity, the $1/r$ potential	Partial, closure open	E-GRV-0041
amplitude-phase	quantum mechanics	Born rule, CHSH, Tsirelson by the discrete route	Parallel	E-QTM-0020, E-CMP-0013
recoverability	thermodynamics, decoherence, horizons	one functional R , three observers	Derived, F	E-GRV-0040
3+1 projection	Navier-Stokes	the fine quadratic conserved exactly, the coarse gradient rises only transiently, no blow-up	Partial, fine exact, coarse transient	E-FLD-0009, E-FLD-0010
cosmic averaging	FLRW, inflation, the spectrum	the wake as a discrete de Sitter phase, the current divergence localized at the growing edge	Background firm, wake divergence shown, spectrum open	E-GMT-0003, E-CSM-0001, E-CSM-0043
spin	circulation quanta	24-cell circulation, the binary tetrahedral double cover acts, a 2π turn negates a spinor	Parallel, double cover shown, F	E-GMT-0026, E-SPN-0042
method	one assignment fits every arena	single-lever rigidity, thresholds set before running	Derived as method, F	E-MTH-0001

The bottom two rungs are Derived on the substrate. Conservation is exact, and its coarse-grained face is the continuity law. These are the firm floor the rest stands on.

The middle rungs are Parallel. Vibe reaches the Born rule, the Tsirelson bound, doubly-special dispersion, area-law gravity, and the $1/r$ potential, but by a discrete route rather than by the continuum's amplitude-phase or thermodynamic argument. We do not claim these as derivations of the continuum statements.

The open rungs are listed below with their falsifiers. Full Lorentz invariance on a fixed lattice is the sharpest. The explicit Einstein-like closure must be derived from coarse-grained tone-stress, not asserted. The Navier-Stokes rung shows both faces, the fine quadratic conserved exactly in E-FLD-0009 and the coarse gradient rising only transiently and recurring in E-FLD-0010, so the reversible lattice has no blow-up at either scale and no Millennium-grade regularity proof is claimed. The variation-to-geometry rung is reached only through a gapless free-fermion proxy, because the bare ternary tone field is short-range. The perturbation spectrum of cosmology is open even where the background expansion and the wake-divergence identity of E-CSM-0043 are firm.

3. The Floor

Continuity

Everything else in the bridge depends on this rung. Until it runs, the claim that the coarse-grained face of the base is a conserved-current field theory is a well-posed conjecture. After it runs, the central identification is a measured fact on the lattice. Coarse-grain the conserved tone-current over growing blocks and its discrete divergence is exactly zero at every block scale. The charge change inside each block equals minus the net flux across its boundary, with no residual. That is the discrete statement of $\nabla \cdot J = 0$, the one law a conserved-current program posits and forces everything else from [42].

Finding 1 On the flat torus (E-GRV-0039) the per-block residual is an integer-exact zero at block scales 1, 2, and 4, with no tolerance allowed. The conserving rule is divergence-free at every coarse-graining. The lossy control injects a per-cell sink and its residual is 2752, 704, and 192 at the same scales, nonzero everywhere and never shrinking. On the genuinely curved $\{3, 4, 3, 4\}$ mesh (E-GRV-0042) the per-region residual is again integer-exact zero at every ball size and every beat over thousands of curved cells, with global conservation and exact reversibility, while the lossy rule breaks it by exactly the charge it destroys inside each ball. Tier F, **Derived**.

The argument is worth walking through, because the identity is so clean it can look like an accident. Every dock carries one tone per direction. From these, build two plain things. The density is the signed sum of the tones, one number saying how much charge sits at the dock. The current is each tone weighted by the direction it points and added, one arrow saying which way the charge flows. That pair is the tone-current, built from what is there. Pick any patch of docks and call it a block. Its boundary is the set of direction links that cross from inside to outside, and each crossing link carries one unit of charge stepping over the edge in one beat.

Now ask why the charge inside changes. The rule does two things and neither makes nor destroys charge. Collide rotates tone among a dock's head-on direction pairs and leaves the dock's total untouched. Stream slides each tone to the neighbor it points at, so a charge that leaves one dock arrives whole at the next.

Nothing evaporates and nothing appears. The only way the charge inside a block can change over a beat is for some charge to physically walk across the boundary. The change inside equals the net amount that crossed the edge exactly, by the way the rule is built. That sentence is the discrete divergence theorem.

Block-averaging carries the per-block balance up to the field equation untouched. Average over blocks of growing size and the per-dock jitter blurs while the density and current settle into smooth fields. The same sum rule, holding at every block and every scale, is exactly the statement that the rate of change of density plus the divergence of the current is zero. That is $\nabla \cdot J = 0$, the same sentence read at a coarser grain. The lossy control is the load-bearing check. Its sink lets charge vanish from a block without crossing the boundary, so the books do not balance and the residual stays nonzero at every scale. Closure comes from conservation. It is not a byproduct of the smoothing that any coarse-graining produces.

So there are not two laws. A continuum program posits continuity at the field scale and forces the rest from it. The base never posits it. The base conserves charge dock by dock, and block-averaging cannot break a balance the rule keeps exactly at every step. Charge conservation at the cell scale is continuity at the field scale, one law seen at two grains. The remaining piece is the explicit map from the lattice sum rule to a single named continuum field with its flow direction and its stress. The base is literally a lattice gas, the canonical engine that produces emergent hydrodynamics from one conserved local rule [50, 8], and the curved run so far streams along the Coxeter generators rather than the full 24-direction coin, so carrying the complete direction set onto the curved mesh is the next step. The conservation is exact because the rule is reversible [21, 9], and the recovery of a smooth four-dimensional continuum from discrete building blocks is the same destination the causal-triangulation program reaches by a different road [20, 28].

The Coin

The substrate is not an arbitrary choice. The cell is forced, and the same structure that forces it is the structure a four-dimensional lattice gas needs to be isotropic. The $\{3, 4, 3, 4\}$ honeycomb is one reflection group, a five-node Coxeter path labeled 3, 4, 3, 4, and its subdiagram signatures force three geometries from one object [17]. The finite end gives the spherical 24-cell as the cell. The affine cubic end gives ordinary three-dimensional space on the cusp horosphere. The bulk is the expanding hyperbolic curvature between them. The cell's 24 directions are the D_4 root system, and they arrive four independent ways: as the two-step tone-words in four slots, as the D_4 roots, as the 24 unit quaternions that form the binary tetrahedral group, and as the four-dimensional kissing number [23]. The binary tetrahedral group is the double cover, so spin is built into the directions from the start.

Finding 2 The 24-cell is selected from two unrelated directions at once. The spinor test requires 24 directions so that matter can carry spin without bolting it on, and the isotropy requirement of a four-dimensional lattice gas independently requires 24 directions so that emergent flow has no preferred axis. E-GMT-0026 confirms the 24-direction coin carries the order-1152 isotropy while the cubic-coin controls at 384 and 48 fail. Tier F, **Derived** for the base. Neither requirement was tuned to the other.

The spinor test is one question, walked through with as little machinery as possible. A spinor is a thing that does not come home after one full turn. Rotate it 360 degrees and it returns flipped, carrying a minus sign, and only a second full turn brings it truly back. Everything ordinary, a scalar or a vector, returns after one turn with a plain plus. Spin one-half is the odd case, and all matter, every electron and quark, is spin

one-half. So the test is just this. Rotate the world once. Does the internal state come back as itself or as its own negative. A cell's tone and its memory are scalars, and scalars do not move under rotation, so they always come back plus. The only place a spinor could hide is the set of directions, because directions do shuffle when you rotate.

For the 24 directions of $\{3, 4, 3, 4\}$ the shuffles sort into three irreducible families. The split is $24 = 8v + 8s + 8c$. The $8v$ is a vector, ordinary, plus after one turn. The $8s$ and $8c$ are the two spinors of $\text{Spin}(8)$, the minus-after-one-turn families. This three-way split is triality, the unique threefold symmetry only D_4 carries, and the 24 directions are exactly the D_4 roots, so they carry it automatically [22]. The spinors are already sitting in the direction set.

The contrast with twelve directions is decisive. The $\{5, 3, 4\}$ honeycomb has 12 directions, and under the icosahedral group they split as $1 + 3 + 3' + 5$. Every piece is whole-number spin and every piece comes back plus. There is no minus anywhere. A plain reshuffle of twelve things cannot hold a spinor, because spinors live only in the double cover, which a permutation can never reach. So $\{5, 3, 4\}$ is a scalar substrate, and spin would have to be added by hand. The 24 directions of $\{3, 4, 3, 4\}$ are the D_4 roots whose cell-group is the binary tetrahedral double cover, where a full turn already acts as minus one. This was computed, and it is the single decisive difference. Everything else tested came out equal for both honeycombs. Spin is the one place the two honeycombs differ.

Now drop spin and ask a separate question. A coarse-grained lattice gas is only physical if its emergent flow keeps no leftover preferred direction, and in four dimensions that demands a specific symmetry. The classical lattice-gas literature already found which one. The face-centered hypercubic lattice, with exactly 24 directions and symmetry of order 1152, is what four-dimensional hydrodynamics needs to come out isotropic, and below that count the emergent flow keeps a grain of the lattice [50, 8]. The 24-cell has exactly 24 directions and that same order-1152 symmetry. E-GMT-0026 confirms the 24-direction coin is isotropic while the cubic controls at 384 and 48 fail.

Two independent requirements meet at one cell. The spinor test knows nothing about fluids. The isotropy requirement knows nothing about spin. One demands 24 directions so that matter can exist without bolting spin on, the other demands 24 directions so that emergent flow has no preferred axis, and they arrive at the same cell from opposite concerns. The cellular-automaton program independently predicts that a deterministic discrete rule should underlie quantum behavior [14], the reversible-rule tradition grounds the rule on the same directions, and the discrete route to relativistic dynamics has a clear precedent [15]. The substrate is the most over-determined choice in the theory [28].

Recoverability

One number ties together results the base reaches separately: entropy, the area law, the boundary screen, and measurement. The recoverability of an observer is R , the accessible structure over the total structure. It is a single currency for three things at once. Thermodynamics is lost recoverability. Decoherence is recoverability flowing into the environment. A horizon is recoverability bleeding across a boundary. The functional is imported from the conserved-current program [42] and realized directly on the substrate.

Finding 3 E-GRV-0040 lays a localized deterministic signal on the $\{3, 4, 3, 4\}$ mesh and reads the same R with three observers. The global observer recovers everything, so $R = 1$ at every beat. The window observer watches R fall toward zero as structure streams across a fixed boundary, the horizon regime. The coarse observer holds R capped under one half as the fine signed current cancels inside blocks, the decoherence regime. A lossy control destroys structure, so even the global observer reads only 0.958. Tier F, **Derived**. One functional read by three observers gives the three faces of recoverability.

Start with what accessible-over-total means. Lay down a structured signal. The total is everything in that signal, the whole of what is really there. The accessible part is how much of it a given observer can reconstruct, given what it is allowed to see and how finely it can look. Recoverability is the accessible part divided by the total, running from zero to one. The key move is that the total never changes. Only what an observer can reach changes, so the same signal yields a different number for a different observer, and that single number is the whole story.

The global observer sees the entire mesh at full resolution and watches every beat. Under the reversible rule it recovers everything, so its recoverability is exactly one, forever. The rule is a reversible permutation that never destroys structure, only moves it from here to there, so nothing the signal carried is ever lost to a watcher who can see all of it. This is conservation read as a number. The window observer can only look through a bounded patch. The signal does not stay put, and beat by beat structure streams out of the window into the region beyond. The observer cannot follow it across the boundary, so its recoverability bleeds toward zero. Nothing was destroyed. The global observer of the first reading still reads one. The structure is simply beyond reach, on the far side of a boundary the windowed observer cannot cross. That is what a horizon is.

The coarse observer can see the whole mesh, but only through a blur. It reads averages over blocks and never the fine signed detail inside a block. The signal carries pluses and minuses sitting side by side inside those blocks, and when the observer averages they cancel, so the fine structure vanishes from its view and its recoverability is capped under a half. The structure is fully present. The global observer still reads one. The coarse observer just cannot resolve what cancels inside its blocks. That is decoherence, a signal present and unreadable at the same time, lost to averaging rather than to destruction.

The lossy control is the discriminator that keeps this clean. Three restricted observers all report loss, and the danger is that loss from restriction looks identical to loss from destruction, in which case the whole picture would prove nothing. So we run a rule that genuinely destroys structure as it goes. Now even the unrestricted global observer cannot read one. Its recoverability drops to 0.958. Under the reversible base only restricted observers lose and the global observer always recovers everything. Under a lossy base the structure itself is gone, so even the all-seeing observer comes up short. A limited observer's loss is lost access. A lossy base's loss is lost structure. The single number tells them apart.

These are not three physics problems forced under one label. They are the same ratio read by observers with three different limits. Conservation is the global reading, a horizon is the windowed reading, and decoherence is the coarse reading. The base did not need a new mechanism for thermodynamics. A reversible conserving rule plus a bounded observer already carries it, and R is the one number that reads it out. This is the discrete, integer-checked counterpart to the thermodynamic-gravity programs that read gravity as an information force on a screen [46, 48, 11], to the entanglement account of emergent geometry [36], and to the observer-as-

boundary framing where classicality is earned rather than free [5]. The coarse-causal-power measures of the effective-information program [10] give a second yardstick on the intervention side, a cousin of recoverability read from how much causal power a coarse layer holds [28].

4. The Climb

Relativity

This is the hardest objection. A fixed deterministic lattice has a rest frame, the frame in which the lattice sits still, and special relativity says there is no such preferred frame. So a fixed-lattice theory looks ruled out from the start. The objection is real, and we do not dismiss it. What the substrate shows so far is a deformed, infrared-restored relativity that is measured at low order and conjectured to be exact, with full Lorentz invariance left open and the obstruction named.

Finding 4 The base carries a finite causal speed and a conserved momentum current with a sharp speed of one (E-RLT-0039), and a doubly-special dispersion that bends with a second invariant and recovers Einstein at low energy (E-RLT-0010), with the Lorentz violation suppressed to high order (E-RLT-0018, E-RLT-0023). These are Tier P, **Parallel**, reached by a discrete route rather than the continuum’s argument. Full exact Lorentz invariance on the fixed lattice is Tier O, **Open**.

The reply rests on two invariants. Ordinary relativity has one invariant everyone agrees on, the speed of light. A doubly-special relativity has two, the speed of light and a fixed length scale, here the lattice spacing. The claim is that all observers agree on both, so the lattice spacing does not single out a frame any more than the speed of light does. A second invariant is a deformation of relativity, not a contradiction of it. This is what E-RLT-0010 measures, a dispersion with a second invariant that returns to the smooth relativistic one at low energy.

The lattice hides at low energy because everything we measure is enormously larger than the lattice spacing. At those scales the discreteness is averaged over so completely that the dispersion, how energy depends on momentum, is the smooth relativistic one to extraordinary precision. The grain shows only in how the dispersion bends, and that bending is pushed to impossibly high energy. So in the infrared the theory looks exactly Lorentz invariant. The load-bearing conjecture behind this is symmetry restoration. The substrate’s exact symmetry is the finite group F_4 , not the continuous rotation group, and the claim is that under coarse-graining the discrete F_4 flows to a continuous symmetry, the way a fine enough polygon is indistinguishable from a circle. That is a flow, not an identity, and it is the open part of the rung.

What is shown and what is owed are kept separate. Shown, and firm where measured: a ballistic light cone with a sharp speed of one, a doubly-special dispersion, and a Lorentz violation suppressed to high order. Owed: exact Lorentz invariance derived from the fixed lattice. The obstruction is well known. A fixed deterministic lattice is standardly expected to pick a preferred frame, and the causal-set program achieves exact Lorentz invariance only by random sprinkling, the move the base forbids [12]. A deterministic discrete rule that can fail to be Lorentz invariant and can fail to look local once coarse-grained is exactly the frontier the cellular-automaton interpretation names [14].

The status is a deformed, infrared-restored relativity measured at low order and conjectured to be exact, with a stated falsifier. There is precedent for relativistic dynamics emerging from a local discrete rule. The

Dirac equation has been derived from a quantum cellular automaton along a doubly-special route, the closest analogue to what the base would need [15], and the iterant construction builds relativistic structure from a discrete sign-alternation kin to the ternary tone and its reversible coin [32]. The caveat is that those automata are quantum and unitary while the base is classical, deterministic, and ternary, so the precedent points the way without transferring directly. The falsifier is straightforward. If the lattice leaves a permanent preferred frame at low order, precision astrophysics would already have seen the Lorentz violation, and it has not [28].

Gravity

Several of the strongest top-down programs reach the same claim. Gravity is the thermodynamics of the entanglement that holds geometry together. Jacobson derived the Einstein equation as an equation of state from horizon entropy [46] and later from vacuum entanglement equilibrium [47]. Padmanabhan read gravity off the counting of surface degrees of freedom [48]. Verlinde made it an entropic force [11]. Van Raamsdonk made the claim literal, that geometric connectivity is entanglement structure, so that cutting the entanglement between two regions tears the space between them apart [36]. Every one of these needs a finite count of states and a microscopic substrate it does not supply. The base supplies exactly that, and reaches the same result by a different route.

Finding 5 On a free-fermion realization whose hopping graph is the bond geometry (E-HLG-0021), dialing one throat bond from open to pinched drives down together both the entanglement entropy across the throat and the geometric connectivity that sews the two halves, with correlation near one, while the throat connectivity collapses below a tenth and a fixed far bond keeps over half of its own. The geometry pinches off exactly where, and only where, the entanglement is removed. In the mediator witness (E-GRV-0041) two regions sharing no direct bond become entangled only through a geometric mediator: their mutual information covaries with the mediator strength at correlation 0.96 and collapses to order 10^{-24} when the mediator is cut, while a direct bypass bond retains 0.997 of the entanglement under the same cut. The co-variation is integer- and correlation-checked with controls that could have failed. The rung as a whole is Tier P, **Parallel:** the entanglement-geometry and thermodynamic-gravity story is reproduced, a full stress tensor sourcing curvature is not.

The throat result is Van Raamsdonk’s picture made spatial and concrete. Split the system into a left half and a right half. The entanglement across the cut is the shared structure neither half holds alone, and the geometric connectivity is how much of the bond web actually bridges the two halves. In the experiment the bonds are also the channels along which the halves come to share structure, so closing the one throat bond reduces both readings at once. They fall together because they are two views of one fact: how strongly the bond ties the halves together. The far bond is the control. If closing one link quietly broke the whole graph, the result would be an artifact. It does not, so the severing of geometry and the loss of entanglement are one event.

The mediator witness is the sharper test, the discrete version of the proposed tabletop experiment where two masses entangle through their mutual gravity [4, 45]. The logic is borrowed from constructor theory: a merely classical go-between cannot entangle two systems that never interact directly. It can correlate them, the way two letters from one sender are correlated, but it cannot entangle them into a single joint state. So if the two regions end up genuinely entangled, and the only path between them was the mediator, the mediator was not classical. The covariance shows it, the severance confirms it, and the bypass bond rules out the

trivial explanation that cutting any bond disrupts everything. The mediating geometry must itself carry quantum structure, and that is read off without ever measuring the mediator.

This is the link the continuum programs assume, now shown on a discrete base. The same base delivers the finite state count Jacobson and Padmanabhan need, and the strict determinism the holographic constructions usually reach only through ensemble averaging. Jacobson, Padmanabhan, Verlinde, and Van Raamsdonk all point at one destination, and the substrate reaches it from below [28].

The open items are listed here. The base reproduces the entanglement-builds-geometry link and the thermodynamic framing, but it does not yet derive an Einstein-like field equation from coarse-grained tone-stress rather than asserting one. Fully reversible curved-bulk gravity is Open. The dark-sector scoreboard, the radial-acceleration relation that must not depend on mass alone, is a real-world constraint the substrate has not yet met. Stated plainly, this rung runs Parallel, and deriving a stress tensor that sources curvature remains the unsolved problem.

The Quantum

The quantum layer is emergent, not part of the base. Raw ternary tones are not amplitudes. The reconstruction programs set, from the top, a checklist any theory claiming to be quantum must pass. Chiribella, D’Ariano, and Perinotti rebuilt quantum theory from informational principles with purification at the center [16, 15]. Hardy reconstructed it from five reasonable axioms [34]. Masanes and Mueller derived it from device-independent requirements [31]. Spekkens showed which features are merely epistemic and which are genuinely non-classical [41]. The task of the base is to pass that checklist from a discrete deterministic substrate, and to supply the microscopic states those selection theorems only assume.

Finding 6 A deterministic base with no injected randomness reproduces the quantum scoreboard: the Born rule with the exponent 2 **Forced** rather than fitted, the classical CHSH bound 2 and the Tsirelson bound $2\sqrt{2}$, contextuality at the Peres-Mermin value 6 against the noncontextual 4, Leggett-Garg at the quantum $3/2$ against the macrorealist 1, the Kochen-Specker structure, purification on the reversible rule (E-QTM-0020), and walk complexity (E-CMP-0013). No amplitude, no imaginary unit, no probability rule, and no Standard Model constant is fed in. The rung is Tier P, **Parallel**. The single definite measurement outcome is Tier O, **Open**.

The surprising part is that the base is deterministic. The textbook intuition is that a deterministic local model is capped at the classical bound of 2, and that saturating the Tsirelson value $2\sqrt{2}$ needs genuine quantum amplitudes. The base reaches that quantum bound from a reversible deterministic permutation with no coin flipped anywhere. It does not break Bell’s theorem. It saturates the exact value Bell’s framework grants a quantum system while being deterministic underneath. The price that keeps this consistent is that the base is not Bell-local. The apparent randomness is ignorance over deterministic states, not fundamental chance. That is why a deterministic base reaching a quantum scoreboard is a real result and not a relabeling.

The distinction between reproduced and derived matters here. Reproduced means the emergent layer exhibits the value, measured on the substrate, matching standard quantum mechanics without being put in. Derived means the value is forced out of the base with no quantum target supplied. Most of the scoreboard is

reproduced. The Born exponent 2 is the one item that is forced, fixed by positivity rather than fitted. So these behaviors emerge from a deterministic base, but they have not each been derived from first principles.

On this base, a measurement is a system relaxing into a single state. The measured system is a small body coupled to a huge environment. Left alone, it genuinely holds more than one possibility at once, a true superposition. When influences arrive the balance tips and the body settles deterministically into one configuration. No special collapse law is added. The only law running is the same reversible rule that runs everywhere, now coupled to a large environment, and decoherence does the work: the overlap between outcomes scatters across countless moving parts and can never be gathered back. The pointer basis is set by how the subsystem couples to the rest during the update, not chosen by hand. The squared weights then give proper Born probabilities and the standard CHSH and Tsirelson values. The continuum version of this story inserts a dissipation rate by hand. The base needs none. Irreversibility emerges from a reversible base coupled to a large environment, and the ever-growing wake supplies the one-way scattering on its own.

The base is superdeterministic in character, not Bell-local. Bell showed that no theory can match the quantum correlations while keeping locality, realism, and statistical independence at once. The base keeps determinism and realism and drops statistical independence: on the mesh the measurement settings and the measured system share a common deterministic past. This is not a conspiracy. The correlation is structural, the ordinary condition of one connected causal history, not a coincidence staged against the experimenter. This is Hossenfelder’s defense made concrete [43]. The base explains why the correlations rise to the Tsirelson value and stop, without restoring local realism. This is also the precise place where the base parts from Christian. The base natively contains his geometric objects: the boundary three-sphere, the octonionic seven-sphere, and the Clifford and spinor structure of the 24-cell. But it refutes his claim that this geometry restores local realism [25].

What stays Open is the single definite outcome. Reproducing the statistics does not, by itself, show from the rule why one outcome occurs rather than another. That derivation is still owed, and it has to come from the substrate on the real mesh rather than a toy model. The stance carries its own falsifier. If a sharpened Bell or contextuality argument ever shows that no deterministic base can reproduce a quantum result the base claims, the determinism commitment fails.

Matter

This is the richest correlation and the sharpest open problem. Many independent programs reach the same exceptional algebra from the top. The base is the only one that builds it from the bottom, a discrete ternary automaton. The shared nucleus sits on one chain, from D_4 through the octonions toward E_8 . The 24-cell is the D_4 root system, its 24 vertices the unit integral octonions [17, 23, 2]. F_4 , the order-1152 cell symmetry, is the automorphism group of the Albert algebra $J_3(\mathbb{O})$ of dimension 27, the cell’s internal state [39]. The Standard Model gauge group descends from the octonions [6, 22], and the weak mixing angle $\sin^2 \theta_W = 3/8$ at unification is the standard grand-unified value [18]. The nucleus is firm. The three generations are open on every side.

Finding 7 One full generation is built, not posited (E-FND-0020), with the 24-cell coin a firm base result (E-GMT-0026). The cell’s 24 directions are the octonion seed, their seven imaginary units acting by left multiplication as an exact seven-generator Clifford algebra. Pairing six generators into three ladder modes

gives eight states with color multiplicities 1, 3, 3, 1 and electric charges 0, 1/3 thrice, 2/3 thrice, and 1, the neutrino, three down quarks, three up quarks, and the electron, with a mirror copy supplying the antiparticles for sixteen states in all. Given the octonion seed the structure is rigid: the Clifford algebra, the three modes, the 1, 3, 3, 1 pattern, and the fractional charges are all **Forced**. This is Tier F for one generation. Three chiral generations are Tier O, **Open**.

The word “Forced” is exact for one generation and must not be stretched further. The 24 directions are the root system of D_4 , the same 24 unit integral octonions, so the nucleus is not added by hand, the cell’s own directions are the octonions [2]. Their left-multiplication matrices square to minus one and anticommute, the defining pattern of a Clifford algebra. This is exact rather than approximate, because what compose are the multiplication matrices, not the non-associative octonions. Climb one level and the cell’s finite symmetry F_4 is the automorphism group of $J_3(\mathbb{O})$, a 27-state internal algebra whose long roots are the 24 directions [39, 22]. Pairing the generators into three modes and setting the electric charge to a third of the filled count delivers one generation’s colors and charges with no free parameter.

The qualification falls on 3/8. The weak mixing angle at unification is a counting fact about the charges above, and the number that appears is the standard grand-unified value of Georgi and Glashow and of $SO(10)$ [18], which the base reproduces from its discrete charge content and which then runs to about 0.231 at the Z mass. The counting is firm once the one-generation charge content is in hand, but 3/8 is a known group-theory value, reproduced and not freshly derived, and the low-energy match uses standard renormalization-group running, which is external physics input. That running is Tier B. It is fed in, and it must never be read as a pure substrate prediction. No Standard Model coupling or mass is derived here. They remain the free parameters the Standard Model already has, each now carrying a geometric origin rather than a value.

The trouble begins at the threefold. D_4 has a unique order-three symmetry, triality, cycling three eight-dimensional pictures written $8v + 8s + 8c$, and it is tempting to read those three as three generations. That reading fails. The $8s$ and $8c$ are the two handednesses of one generation and $8v$ is the vector that carries the gauge force, so the natural triple is a vector plus two chiralities, not three matter families (E-SPN-0041). A cleaner threefold lives in the Albert algebra, whose frame has exactly three primitive slots, three because the Jordan rule holds for octonionic Hermitian matrices up to size three and breaks at size four where non-associativity sets in. The counting is forced with size four as the control that must fail, and it does. But identifying those three algebraic slots with the three observed generations is Boyle’s conjecture, and it is unproven [29].

The mass hierarchy is a recorded negative (E-SPN-0039). If the three slots were the three generations their masses should split widely, and the base supplies no mechanism. The three slots carry an exact threefold family symmetry, a symmetric vacuum keeps all three masses equal, and the two asymmetric vacua that do split them give two different ratios with no preferred choice, so the splitting carries no geometric information. The mass hierarchy is a free Yukawa input, not a substrate output, because deterministic dynamics cannot break an exact symmetry on its own. For this base Boyle’s conjecture resolves negative on the hierarchy (E-SPN-0016). The continuum embeddings that try to fit three generations into E_8 hit the Distler-Garibaldi no-go, which forces mirror fermions nature does not contain and which has stood for over fifteen years [19]. The only conceivable escape is that the no-go is a continuum theorem while the base is discrete, and that has not been shown.

The status is stated plainly. Two real, forced threefolds exist on the substrate, triality of order three and the Albert algebra as three times nine, and their identification with the three observed generations is Open on both sides. The naive dimensional match fails. This is a single wall, the same wall Furey, Lisi, Dixon, and Dray-Manogue attack from the continuum side [6, 22]. Triality having the right count is necessary, not sufficient, and the base does not imply a derivation it has not made. One generation is compelling and forced. Three chiral generations are not derived.

Fluids

The base rule is literally a lattice gas. Collide rotates tone among a dock's head-on direction pairs and conserves the dock total, then stream slides each tone to the neighbor it points at. That is the exact shape of the discrete machine the classical literature built to recover fluid behavior at coarse grain [50]. So hydrodynamics is expected from the substrate rather than an added assumption. The grade for this rung is **parallel**: the engine is the right one and one cosmological scalar is measured, while a full continuum fluid with a proven regularity bound is not claimed.

The lattice-gas lineage imposes a sharp geometric requirement. A coarse-grained lattice gas is only physical if its emergent flow keeps no leftover preferred direction, and in four dimensions that isotropy needs a specific symmetry. The face-centered hypercubic lattice gas of the classical work needs exactly the 24-direction coin whose symmetry group has order 1152 [8]. That is vibe's 24-cell, selected on independent grounds. The structure a four-dimensional fluid needs to come out isotropic is the same structure the base already carries.

Finding 8 The coarse-grained tone-current carries a hydrodynamic gradient energy, and the wake is a measured expansion scalar. The Navier-Stokes rung is partial: the fine bound is exact and the coarse gradient steepens only transiently, with no sustained blow-up.

The conserved tone-current is a density and a current at every dock, the same pair that obeys continuity. At coarse grain it carries a velocity-gradient energy, the quantity whose possible runaway growth is the open question for the continuum equations. Two experiments read this quantity at the two grains. **E-FLD-0009** shows an enstrophy-like quadratic conserved exactly by the reversible knit, the fine lattice bound that a continuum truncation forgets. The lossy control collapses it, so the bound is caused by reversible conservation and not by generic smoothing.

The coarse-grained result is reported as measured. **E-FLD-0010** measures the coarse velocity-gradient energy and finds it rises above its start by a transient steepening of about 11 percent, then recurs rather than running away. There is no sustained blow-up at either scale. The fine quadratic is held exactly and the coarse gradient only flexes and returns. This is a partial result: a Millennium-grade regularity proof is neither needed nor claimed, and the absence of a blow-up on the reversible lattice is itself the finding.

The cosmological scalar is firmer. **E-CSM-0043** measures the divergence of the current on the growing mesh and finds it equals the count of newly born cells at the edge, positive and growing with the shell counts, while it is integer-exact zero in the settled bulk. The static-mesh control shows no divergence, so the divergence is sourced only by the growth of the mesh. This moves the wake's role as an expansion term from an assertion to a measured continuity identity, a hydrodynamic expansion scalar at the boundary where the mesh grows.

What remains is the link to lab physics. The target for the coarse-graining is relativistic hydrodynamics built from one conserved current, the kind of emergent fluid the lattice-gas program was designed to reach. That explicit map, for a single named continuum field with its flow direction and stress, is not built here. Fluids stays parallel: the discrete engine matches the canonical one, the fine bound and the wake scalar are measured, and the continuum regularity is left open [28].

Scale

The whole upward program assumes the base coarse-grains into a tower of faithful layers, each with a law of its own, stacked from the cell up to the continuum. Coarse-graining is the single bridge that builds the tower. Group the cells into blocks, read one summary value off each block, and ask how those summaries change from beat to beat. If the blurred picture has a law of its own that does not secretly depend on the fine detail thrown away, the layer is real rather than a convenient label. The question for the whole tower is whether those laws hold at every level without contradicting each other.

The precise test of a single rung is the commuting square. From a micro state there are two routes to the coarse state one beat later. Run the real rule one beat and then coarse-grain, or coarse-grain first and then run the effective rule on the summaries. When the two routes land on the same corner, the coarse layer is closed and faithful. When they disagree, the layer leaks and keeps needing the fine grain to predict itself. Whether the full tower is self-similar, with one effective rule recurring step after step and a single geometric ratio relating each layer to the next, is E-DEC-03, ready to build. The candidate ratio is the per-shell warp factor near 18.28, an exact Perron number forced by the honeycomb, with the shell counts 1, 24, 456, 8376 exact. A flat Euclidean control is the discriminator, since coarse-graining cleans up almost any substrate a little and the result only counts if the flat run fails to clean up while the curved one succeeds. The yardstick for whether each rung is causally real is effective information per layer [10], the measure of when a coarse account becomes a sharper cause than the level below it. The tower is one more discrete-to-continuum route of the kind the computational-universe and cellular-automaton programs explore [44, 21, 13].

Finding 9 One assignment of the shared substrate levers passes every arena. Pulling a single lever drops a whole cluster of arenas at once, so the theory is rigid rather than a set of separately tuned fits.

The discipline that keeps the rungs from contradicting each other is the consistency matrix. Put the arenas down the side and the few substrate parameters across the top, fix one assignment of those parameters, and derive every arena from that single assignment. There is no private column of knobs per arena. The same two levers, the 3,4,3,4 frame with its 24-direction coin and the conserving reversible rule, feed every cell of the table. E-MTH-0001 runs four arenas under three assignments with the refutation thresholds fixed in advance.

Assignment	Conservation	Reversibility	Isotropy	Continuity	Total
{3,4,3,4} frame, conserving rule	pass	pass	pass	pass	4 of 4
Lossy rule, same frame	fail	fail	pass	fail	1 of 4
Cubic 6-direction frame, same rule	pass	pass	fail	fail	2 of 4

Each off-assignment breaks a cluster of cells at once rather than just one. Swap in the lossy erasing rule and conservation, reversibility, and continuity fall together, because all three hang on the same conserved property, so the matrix passes only one arena. Swap in the cubic six-direction coin and isotropy fails, since

the 24-cell zeroes a fourth-moment anisotropy that the cubic frame leaves stuck, and the four-dimensional continuity closure is undefined on a three-dimensional frame, so that arena falls too. Only the committed assignment closes the matrix. This is Tier F, **derived**.

The single-lever structure is what makes the theory falsifiable as a whole. Because the levers are shared, no failing arena can be repaired without risking another, and there is no per-result dial to absorb a bad fit. A single real disagreement would not just dent one corner. It would force a shared lever and drag a whole cluster of currently passing arenas down with it. That is the opposite of a collection of separate fits, each with its own knob, where a bad arena is patched in isolation. This first matrix has four firm arenas. The fuller matrix the program aims at adds the open arenas, where the refutation conditions are stated but the cells are not yet green: exact Lorentz invariance on a fixed lattice, the explicit gravity closure, three chiral generations, and serves as causally real. The discipline is the same at every rung. State the thresholds first, run, and report the result, pass or recorded negative.

5. Selves

Selves

This is the most speculative material in the paper, and it begins with a firm caveat. Raw vibes are not physics. The ternary tones are values, not particles, and the substrate does not map onto the world in one step. Between the base and any physics there are one to three emergent middle layers, and selves, where they appear at all, live in that middle and not in the base. Everything below is labeled by tier, and where a claim is conjecture it is marked **open**, Tier O.

The upper program rests on one constraint that doubles as a prediction. A closed, deterministic, reversible base sits at the causal-reduction end, where the finest micro account stays the best account and no coarse summary improves on it. Causal emergence, where a coarse layer becomes a sharper cause than the cells beneath it, needs noise or degeneracy to average away, and a reversible rule has neither. So autonomy can enter only through one of two doorways: lossy coarse-graining, where a many-to-one projection injects effective noise, or open subsystems trading tone with the ever-growing wake. The yardstick for whether a layer is causally real is effective information, how much intervening on a cause pins down the effect [10]. This much is Tier F.

Finding 10 Selves cannot come from the closed reversible base. They can arise only in the lossy middle. The pure base churns and makes no bound body, which is the recorded negative the constraint predicts.

The closed-base case was run and reported. At over a million docks, with charge and momentum conserved exactly and the run bit-identical on two machines, a proto-self does not bind. It spreads ballistically at about 1.3 cells per beat. The bare rule delivers identity and radiation, not a bound body. This is the closed base behaving exactly as the constraint says it must, a measured negative tracked in the selves arena, the E-SLF experiments under the decisive labels E-DEC-02 and E-DEC-06. What stays open is that a bound, self-repairing body needs one more discrete ingredient, an attraction, which the bare atoms do not supply. Tier O.

The right test for a candidate self is a body-hit: perturb the structure's own charges and watch its body. Binding has to be confirmed by spatial localization, a pattern that stays put and relaxes back when pushed,

and not by a recurring scalar, because a recurring number is exactly what a simple gas already produces without binding anything. A real self should register as an irreducible high-order cause whose interventions carry more effective information than its own cells do. The decisive run computes that quantity in three cases, the closed base, a lossy coarse-graining, and an open subsystem, expecting selves to appear only in the last two. This is the E-SLF arena, with the controls that the closed base shows reduction and a scramble destroys the structure.

The middle is also where freedom could appear. The claim here is deliberately limited. Two things both called open must be kept apart. Reality-openness asks whether the same total state could yield two different futures, and on a deterministic reversible rule the answer is no, always. Agent-openness asks whether a self holds a picture of alternatives and cannot read its own next act until it settles, and there the answer can be yes. Freedom, on this reading, is the product of modeled openness and self-unreadability, a quantity that is zero for a bare tone with nothing to model, zero for the all-containing whole with nothing outside to model, and large for a part modeling a world bigger than its view. Determinism holds underneath, unchanged. This framing draws on Friston’s Markov blanket read as a Lyapunov-settling agent with no literal inference [26] and on the self as a self-maintaining predicting boundary [5]. The framing of laws as which transformations are possible and impossible is the admissibility lens [7, 4]. All of this is Tier O, the interpretive layer, stated and clearly labeled rather than derived.

The measurable parts of this section are firm. The deepest claim remains a conjecture, Tier O. Conservation is the precondition that lets a pattern settle into an attractor and persist, which is necessary for a self but not yet sufficient, since it does not by itself pull the pattern into a bound body. The model does not prove free will. It defines freedom as two measurable quantities, shows the product should peak in the middle for reasons that trace to the firm base, and predicts the one place a living self could ever come from while ruling out the place it cannot. The identification of the tones with felt pain, peace, and pleasure is the monist reading. It is the one place the model reaches past physics, and it is offered tentatively, as the base axiom seen from the inside rather than a derived result.

6. The Reckoning

The Map

The bridge is real, and it is the strongest correlation in the corpus. Most of the spine is now measured rather than asserted. Some of it runs parallel to independent programs by a different route. A short list of walls stays open, and they are shared with everyone working this problem. The map below sorts every result into one of three tiers. **Firm** means established on the substrate, with a control that could have failed, graded Derived. **Parallel** means the same destination reached by an independent route, not a derivation of the continuum statement, graded Parallel. **Open** means a well-posed conjecture with a stated falsifying test, unsolved on both sides, graded Open.

The firm tier rests on exact, checkable results. The conserved current and its exact continuity closure hold as integer-exact identities, on the flat torus and on the genuinely curved $\{3, 4, 3, 4\}$ mesh alike. One recoverability functional carries horizons, decoherence, and conservation together. Two regions entangle only through the geometric mediator. The order-1152 coin gives a four-dimensional lattice gas the isotropy it needs, where cubic

coins fail. The 24 directions act as the binary tetrahedral double cover, so a full turn negates a spinor. One substrate assignment passes every arena, and perturbing one lever breaks a cluster. These are measurements on an explicit object.

The middle tier runs parallel. Vibe reaches the Born rule and the Tsirelson bound from a deterministic base with no injected randomness, instantiates spin as concrete 24-cell circulation, builds geometry from entanglement on the substrate, and supplies the finite state count the thermodynamic gravity arguments need for free. These are the same results reached by an independent route, not derivations of the continuum statements they echo.

The open tier is listed below, each wall with its falsifier. The next section walks through them one by one.

Firm (Derived)	Parallel (Parallel)	Open (Open)
Conserved tone-current, lossy control fails (E-FND-0030)	Born rule and Tsirelson bound by the discrete route (E-QTM-0020, E-CMP-0013)	Exact Lorentz invariance on a fixed lattice [33, 14]
Coarse continuity $\nabla \cdot J = 0$, flat and curved (E-GRV-0039, E-GRV-0042)	Spin as 24-cell circulation, instantiated concretely	The explicit Einstein-like closure from coarse-grained tone-stress
Recoverability functional, three observers (E-GRV-0040)	Geometry from entanglement, paralleling Van Raamsdonk [36]	Triality to three chiral generations [19, 29]
Gravity-induced entanglement through the mediator (E-GRV-0041, E-HLG-0021)	Gravity from thermodynamics, the finite state count supplied [46, 48]	The single measurement outcome
Single-lever consistency matrix (E-MTH-0001)	The dark sector as emergent, paralleling Verlinde [11]	Selves as causally real in the lossy middle
Native isotropy, order-1152 coin (E-GMT-0026)	Doubly-special dispersion, Einstein recovered at low energy (E-RLT-0010)	Emergent locality at scale
Spinor double cover on the coin (E-SPN-0042)	The wake as a measured expansion scalar (E-CSM-0043)	A full continuum field theory with a stress tensor sourcing curvature
Conserved momentum current, sharp speed one (E-RLT-0039)		
The shared algebra nucleus, D_4 , the 24-cell, octonions, E_8 , $\sin^2 \theta_W = 3/8$		

Table 3: The whole-paper scoreboard, by tier.

The bridge stands on its firm anchors. Completing it means turning the parallel results into measured ones. The wake as a Chronoflux expansion scalar is already done. The lattice fluid bound and geometry-dependent settling are still owed, with the Dirac fluid the largest single undertaking. The open walls stand or fall on three tests: exact Lorentz invariance, the escape to three generations, and the causal reality of selves. The reader should be convinced by these experiments hitting their stated thresholds, not by anyone's authority.

Candor

The central claim is a conjecture. It is not yet a completed result. Everything in this paper is an in-silico computation on the substrate rather than a measurement against nature. The gaps are stated plainly here, one short paragraph each, so they are not buried under the firm results.

A field theory is not yet built. The exact result is that the discrete divergence of the conserved charge is zero. A full conserved-current field theory needs a named continuum field, an action, a stress tensor, and equations of motion. The map from the lattice sum rule to a specific field with a stress tensor that sources curvature is unbuilt. Continuity is shown, but the field theory it implies has not been built.

Exact Lorentz invariance on a fixed lattice. This is the hardest open problem. A fixed deterministic lattice is standardly expected to pick a preferred frame, the frame in which it sits still. Causal sets reach exact Lorentz invariance only by random sprinkling, the move vibe forbids [33, 12]. Vibe’s reply is a doubly-special relativity with two invariants, the speed of light and the lattice spacing, and a discrete F_4 that restores to a continuous symmetry in the infrared. Shown so far: a ballistic light cone, a conserved momentum current with sharp speed one, and a doubly-special dispersion that recovers Einstein at low energy. Still open: full exact Lorentz from the fixed lattice. If the lattice left a permanent preferred frame at low order, precision astrophysics would already have seen the Lorentz violation, and it has not [14].

Three chiral generations. Two forced threefolds exist on the substrate, but their identification with the three observed generations fails the naive match. The Distler-Garibaldi no-go forbids fitting three chiral generations into E_8 in the continuum without mirror fermions that nature lacks, and it constrains every algebra-spine program alike [19, 29]. Vibe’s only structural difference is discreteness, which may evade the continuum no-go but has not been shown to. The naive triality reading gives a vector plus two chiralities, a negative result. The escape is open.

The single measurement outcome. Vibe reproduces the Born rule and the Tsirelson bound from a deterministic base with no injected randomness, and every emergent mixed state is the marginal of a pure reversible whole. What it does not deliver is why a single definite outcome is recorded on one run. The measurement step is reached as statistics, not as the selection of one result. The single-outcome problem is open here, as it is everywhere.

Selves as causally real in the lossy middle. The closed reversible base favors causal reduction, so selves must come from lossy coarse-graining or open subsystems. Whether vibe’s selves are irreducible high-order causes, in the sense of causal emergence [10], is unmeasured. The bound self-repairing body still needs one discrete ingredient that has not been isolated. Open.

Emergent locality at scale. A coarse-grained theory is only physical if its effective interactions stay local once averaged [49]. That the substrate keeps locality at scale, rather than developing the disordered nonlocality a graph theory can suffer, is a well-posed test that has not been run to completion. Open.

The other side of the bridge is itself unconfirmed. The continuum program vibe is measured against is unreviewed, unreplicated, and single-author [42]. The bridge takes its principles, targets, and methods, and leaves the rest, but a parallel to an unconfirmed program is a weaker anchor than a parallel to an established one. This is stated plainly.

A page claiming every problem is solved would be worth less than this list of open ones. Each open problem narrows the search and points to what to test next. The thesis is a conjecture that the firm results support but do not prove.

The End

This paper positioned a discrete, deterministic, reversible base against the continuum programs in one disciplined frame, and showed, one result at a time, where the base derives a continuum result, where it runs parallel to a program by a different route, and where the bridge is a well-posed open conjecture with a stated falsifying test. The firm floor is exact and integer-checked. Charge conservation is the continuity law $\nabla \cdot J = 0$, shown on a flat torus and on the genuinely curved $\{3, 4, 3, 4\}$ mesh. The 24-cell coin is selected by a spinor test and carries the isotropy a four-dimensional lattice gas needs. One recoverability functional carries horizons, decoherence, and conservation together. The middle rungs run parallel, and the open walls are listed with their tests.

The conjecture is this. The coarse-grained face of one growing mesh of one substance, carrying a ternary value, moved by one local reversible charge-conserving rule, is a conserved-current field theory of the kind the continuum programs describe. That claim is not proven. It is supported by a firm anchor and a clear list of unbuilt steps, and it stands or falls on experiments hitting thresholds fixed before they ran.

This is exploratory, feasibility-stage work, built to be tested and falsified. The whole comparison rests on a reproducible suite of deterministic, controlled experiments, each paired with a control that would break it if the result were an artifact [28]. The invitation is to run them, to push the parallel rungs toward measured ones, and to test the open walls against nature. A theory should derive what it can, state plainly what it cannot, and remain open to being shown wrong.

7. Appendices

The Experiments

Every result in this paper is checked by a reproducible computation on the substrate, never a measurement against nature. The full suite is a finite, deterministic, runnable battery [28]. This appendix explains the code scheme, the depth grades, the counts, and indexes the key experiments cited in the paper.

The code scheme

Each experiment carries a stable code, registered once in the catalog, of the form E-XXX-NNNN. The XXX is a three-letter arena that groups by topic, GRV for gravity, RLT for relativity, SPN for spin, FLD for fluid, CSM for cosmology, MTH for method, FND for foundations, and so on. The NNNN is a four-digit number, unique within the arena. The four anchors of this paper are E-GRV-0039, E-GRV-0040, E-GRV-0041, and E-MTH-0001. Open frontiers are tracked by a separate E-DEC planning label, which marks an arena's decisive test rather than a catalog entry.

The depth grades

Each experiment is graded by what it actually establishes, from L0 to L3.

grade	what it establishes
L3	an emergent, novel result from the base rule, with a control where the answer should come out no
L2	known physics reproduced on the substrate
L1	known math confirmed
L0	a circular consistency note, kept and labeled as such

The counts

The evidence base is 615 experiments across 18 categories. 84 sit at L3, 389 at L2, 128 at L1, and 14 at L0. Of these, 412 back a specific claim in the papers. A result counts only as the gap between the run and its control. Pass conditions are fixed before running. Where possible they are exact integer or ratio equalities with no tolerance, so a result is an identity rather than a fit. A run is bit-identical on two machines.

The key experiments

The experiments cited in this paper, by code, with depth grade and verdict.

code	title	depth	verdict
E-FND-0030	conservative reversible logic	L2	charge conserved exactly, lossy control fails
E-GMT-0026	the 24-cell coin	L2	order-1152 F_4 isotropy, cubic coins at 384 and 48 fail
E-RLT-0010	doubly-special dispersion	L2	dispersion bends with a second invariant, recovers Einstein at low energy
E-RLT-0039	momentum current and cone	L2	momentum current conserved exactly, sharp speed one, lossy loses it
E-HLG-0021	entanglement-geometry co-variation	L2	connectivity tracks entanglement across a throat
E-QTM-0020	purification	L2	every emergent mixed state is the marginal of a pure reversible whole
E-CMP-0013	walk complexity	L2	the complexity signature of the quantum walk on the substrate
E-GRV-0039	coarse continuity closure	L2	$\nabla \cdot J = 0$ exact at every block scale, lossy nonzero

code	title	depth	verdict
E-GRV-0042	curved-mesh continuity	L2	$\nabla \cdot J = 0$ exact at every region on the curved mesh, lossy breaks it by the destroyed charge
E-GRV-0040	the recoverability functional	L2	one R carries horizons, decoherence, and conservation, lossy global R falls below 1
E-GRV-0041	gravity-induced entanglement	L2	two regions entangle only through the mediator, severed kills it, bypass retains
E-SPN-0042	spinor double cover	L2	the 24 directions act as the binary tetrahedral double cover, a 2π turn negates a spinor
E-FLD-0009	enstrophy bound	L2	an enstrophy-like quadratic conserved exactly, lossy collapses it
E-FLD-0010	coarse runaway	L2	coarse gradient energy rises transiently while the fine bound is exact, no blow-up
E-CSM-0043	wake as expansion scalar	L2	the current divergence equals the count of newly born edge cells, zero in the bulk, dead under the static control
E-MTH-0001	the consistency matrix	L2	one assignment passes 4 of 4 arenas, the lossy rule 1 of 4, the cubic frame 2 of 4

A handful of established results are reproduced across several experiments and cited without a single code: the area law, the $1/r$ potential, the Born rule, CHSH and the Tsirelson bound, the contextuality values, and fermion universality. Two attempts in this round did not build cleanly and are reported as such. A metric from the bare tone correlations failed because the tone field is short-range. A clean geometry-dependent settling time failed because the observable is too noisy. A negative result is itself a finding.

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